

most extensive method + theorem collection in olympiad maths

ONYQZ

19.10.2024

The following is a collection of many different theorems, lemmas, methods and principles you might common across in olympiad mathematics. The aim is to give an overview of what one might need to learn.

This is of course not 100% (but $\approx 99\%$) complete. Moreover, key methods and principles can be used in different fields, in which case I will put them into the section "General". The topics in one subsection are also ranked on **usefulness**, **difficulty of learning and applying** the method (from the perspective of a beginner or an intermediate) and **starting from which round** it is used.

Each category can get 1 to 4 stars.

1 ★: not very useful, easy or regional round resp.

2 ★: somewhat useful, medium difficult or state/province round resp.

3 ★: useful, hard or national round resp.

4 ★: very useful, very hard or IMO/TST level resp.

Since difficulty might vary from country to country, I am just going to do that from my point of view and considering how it is in Germany. Many theorems do not really find an application because they are to niche or difficult for high school competitions. Nonetheless, I wanted to include some of them.

Also note that I cannot and will not go into detail on every topic. There are a lot of "random" lemmas that are useful in some cases but don't have a name. These have to be learned over time or you are just able to derive them on the fly. If you want to study the topics in depth, then please see the resources I recommend in the last section.

Contents

1	General	3
2	Algebra	4
2.1	identities	4
2.2	Inequalities	4
2.2.1	Algebraic Inequalities	4
2.2.2	Geometric Inequalities	6
2.3	Functional Equations	6
2.4	Functions and Polynomials	7
2.5	other results and methods	8
3	Number Theory	10
3.1	classical Number Theory	10
3.1.1	Modular Arithmetic	11
3.1.2	Diophantine equations	12
3.2	Analytic Number Theory	13
3.3	Algebraic Number Theory	14
3.4	other results and methods	14
4	Geometry	16
4.1	Plane Geometry	16
4.2	Solid Geometry	19
4.3	Analytic Geometry	19
5	Combinatorics	21
5.1	basic methods	21
5.2	Graph Theory	21
5.3	Sets	23
5.4	Games, Tiling, Algorithms	24
5.5	other results and methods	24
6	Probability	25
7	Linear Algebra	25
8	resources for self-study	26
8.1	books	26
8.2	internet and blogs	27
8.3	handouts and articles	27

1 General

1. bounding (everywhere)
(round: 1 ★, usefulness: 4 ★, difficulty: 1-4★)
2. extremal principle (esp. combi and graph theory)
(round: 1 ★, usefulness: 4 ★, difficulty: 2-4 ★)
3. proof by contradiction (esp. everywhere except geo)
(round: 1 ★, usefulness: 4 ★, difficulty: 1-4 ★)
4. direct construction (esp. combi, NT)
(round: 1 ★, usefulness: 4 ★, difficulty: 1-4 ★)
5. exploiting symmetry (esp. algebra, geo)
(round: 1 ★, usefulness: 3 ★, difficulty: 1-2.5 ★)
6. wishful thinking + working backwards (esp. algebra, combi)
(round: 1 ★, usefulness: 4 ★, difficulty: 1-4 ★)
7. analyze appearing numbers, functions, conditions, constraints ... (esp. algebra, NT)
(round: 1 ★, usefulness: 4 ★, difficulty: 1-4 ★)
8. partitioning problem into smaller problems/steps (esp. combi)
(round: 2 ★, usefulness: 4 ★, difficulty: 1-4 ★)
9. induction (esp. combi, NT, algebra)
(round: 2 ★, usefulness: 4 ★, difficulty: 2-4 ★)
10. local approach, global approach (everywhere)
(round: 2 ★, usefulness: 4 ★, difficulty: 2-4 ★)

2 Algebra

2.1 identities

1. trigonometric identities
(round: 1 ★, usefulness: 2.5 ★, difficulty: 1 ★)
2. Diophantus' identity + Lagrange's identity
(round: 2 ★, usefulness: 1.5 ★, difficulty: 1 ★)
3. Sophie-Germain's identity
(round: ? ★, usefulness: 1.5 ★, difficulty: 1 ★)
4. Lagrange's identity
(round: ? ★, usefulness: 1.5 ★, difficulty: 2 ★)

2.2 Inequalities

2.2.1 Algebraic Inequalities

1. triangle inequality
(round: 1 ★, usefulness: 2 ★, difficulty: 1 ★)
2. substitutions (uvw, pqr method), transformations, homogenize, expansion (Buffalo Way method), symmetrize, normalize
(round: 1 ★, usefulness: 4 ★, difficulty: 1-3 ★)
3. AM-GM-HM inequality
(round: 1 ★, usefulness: 4 ★, difficulty: 1 ★)
4. power mean inequality
(round: 1 ★, usefulness: 3 ★, difficulty: 1 ★)
5. rearrangement inequality
(round: 1 ★, usefulness: 3 ★, difficulty: 1 ★)
6. differentiate
(round: 1 ★, usefulness: 1.5 ★, difficulty: 1 ★)
7. Cauchy-Schwarz inequality
(round: 1 ★, usefulness: 4 ★, difficulty: 1-3 ★)
8. Titu's Lemma
(round: 1 ★, usefulness: 4 ★, difficulty: 2 ★)
9. Bernoulli's inequality
(round: ? ★, usefulness: 1.5 ★, difficulty: 1 ★)
10. Nesbitt's inequality
(round: ? ★, usefulness: 1.5 ★, difficulty: 1 ★)
11. Radon's inequality
(round: ? ★, usefulness: 2.5 ★, difficulty: 1.5 ★)
12. Turkevich's inequality
(round: ? ★, usefulness: 2 ★, difficulty: 1.5 ★)

13. logarithmic inequalities
(round: 2 ★, usefulness: 3 ★, difficulty: 2 ★)
14. Muirhead's inequality
(round: 2 ★, usefulness: 3.5 ★, difficulty: 2.5 ★)
15. Chebyshev's inequality
(round: 2 ★, usefulness: 3 ★, difficulty: 1.5 ★)
16. Schur's inequality
(round: 2 ★, usefulness: 3.5 ★, difficulty: 1-3 ★)
17. Tangent line method
(round: 2-3 ★, usefulness: 3 ★, difficulty: 2.5 ★)
18. convex and concave functions
(round: 2-3 ★, usefulness: 3.5 ★, difficulty: 2 ★)
19. Popoviciu's inequality
(round: ? ★, usefulness: 2 ★, difficulty: 2 ★)
20. Hlawka's inequality
(round: ? ★, usefulness: 1 ★, difficulty: 2 ★)
21. Weierstraß' extreme value Theorem
(round: 2-3 ★, usefulness: 3 ★, difficulty: 2 ★)
22. Jensen's inequality
(round: 3 ★, usefulness: 3.5 ★, difficulty: 2.5 ★)
23. Karamata's inequality
(round: 3 ★, usefulness: 3 ★, difficulty: 2.5 ★)
24. Hölder's inequality
(round: 3 ★, usefulness: 3 ★, difficulty: 2-3 ★)
25. Langrange Multiplier's method (sometimes "banned", be careful!)
(round: 3 ★, usefulness: 2.5 ★, difficulty: 3.5 ★)
26. Minkowski's inequality
(round: ? ★, usefulness: 2 ★, difficulty: 2-3 ★)
27. Hardy's inequalities
(round: ? ★, usefulness: 2 ★, difficulty: 3 ★)
28. Maclaurin's inequality
(round: 3-4 ★, usefulness: 2 ★, difficulty: 2.5 ★)
29. Newton's inequality
(round: 3-4 ★, usefulness: 2 ★, difficulty: 2.5 ★)
30. Ky-Fan's inequality
(round: 3-4 ★, usefulness: 2 ★, difficulty: 3 ★)
31. Supersums of Square-Weights method, Chinese Dumbass Notation
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 2.5 ★)

32. Steffensen's inequality
(round: 4 ★, usefulness: 1.5 ★, difficulty: 3 ★)
33. Bacteria Method (undefined coefficients method)
(round: ? ★, usefulness: 1.5 ★, difficulty: 3.5 ★)
34. $n - 1$ Equal Value Principle
(round: ? ★, usefulness: 1 ★, difficulty: 4 ★)
35. Vasc's Right Convex Function Method and Left Convex Function Method
(round: ? ★, usefulness: 1 ★, difficulty: 4 ★)

2.2.2 Geometric Inequalities

1. triangle inequality
(round: 1 ★, usefulness: 3 ★, difficulty: 1 ★)
2. Ravi substitution
(round: 2 ★, usefulness: 4 ★, difficulty: 1.5 ★)
3. trigonometric inequalities
(round: 2 ★, usefulness: 3 ★, difficulty: 2-3 ★)
4. r-s-R inequality (extended Euler's inequality) ($2r \leq \frac{s}{3\sqrt{3}} \leq R$)
(round: 2 ★, usefulness: 3.5 ★, difficulty: 1.5 ★)
5. Leibniz' inequality
(round: 2 ★, usefulness: 2 ★, difficulty: 1.5 ★)
6. Finsler-Hadwiger's inequality (+ Weitzenböck's inequality)
(round: ? ★, usefulness: 2 ★, difficulty: 1.5 ★)
7. Erdos-Mordell's inequality
(round: 2 ★, usefulness: 2 ★, difficulty: 1.5 ★)
8. minimization of distance
(round: 2 ★, usefulness: 3 ★, difficulty: 2 ★)

2.3 Functional Equations

1. Cauchy's functional equation
(round: 1 ★, usefulness: 4 ★, difficulty: 1.5 ★)
2. Cauchy's exponential functional equation
(round: 1-2 ★, usefulness: 3 ★, difficulty: 1.5 ★)
3. Jensen's functional equation
(round: 2 ★, usefulness: 2 ★, difficulty: 1.5 ★)
4. Pexider's functional equation
(round: ? ★, usefulness: 1.5 ★, difficulty: 1.5 ★)
5. exploit symmetry
(round: 2 ★, usefulness: 3 ★, difficulty: 2 ★)
6. injectivity, surjectivity, bijectivity
(round: 2 ★, usefulness: 4 ★, difficulty: 1.5-3.5 ★)

7. pointwise value trap
(round: 2-3 ★, usefulness: - ★, difficulty: 3 ★)
8. preimage, orbit, period, fixed point, functional graph
(round: 3 ★, usefulness: 3 ★, difficulty: 3 ★)
9. fff method
(round: 3 ★, usefulness: 3 ★, difficulty: 2.5 ★)
10. forced cancellation method
(round: 3 ★, usefulness: 3 ★, difficulty: 2.5 ★)
11. Squeeze Theorem
(round: 3 ★, usefulness: 2.5 ★, difficulty: 3 ★)
12. supremum, infimum, limit superior, limit inferior
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 3 ★)
13. monstrous functional equations
(round: 3-4 ★, usefulness: - ★, difficulty: 4 ★)

2.4 Functions and Polynomials

1. Vieta's formulas
(round: 1 ★, usefulness: 3 ★, difficulty: 1 ★)
2. Euclidean Algorithm, polynomial long division, Horner's method
(round: 1 ★, usefulness: 3 ★, difficulty: 1-3 ★)
3. intermediate value theorem
(round: 1 ★, usefulness: 4 ★, difficulty: 1 ★)
4. $a - b \mid P(a) - P(b)$ lemma
(round: 2 ★, usefulness: 3.5 ★, difficulty: 1.5 ★)
5. elementary symmetric polynomials
(round: 3 ★, usefulness: 3 ★, difficulty: 2.5 ★)
6. Newton's identities/sums
(round: 3 ★, usefulness: 2.5 ★, difficulty: 2.5 ★)
7. roots of unity
(round: 3 ★, usefulness: 4 ★, difficulty: 2.5-4 ★)
8. Lagrange's Theorem
(round: 3 ★, usefulness: 3 ★, difficulty: 2 ★)
9. Schur's Theorem
(round: 3 ★, usefulness: 3 ★, difficulty: 2 ★)
10. Taylor expansion
(round: 3 ★, usefulness: 3 ★, difficulty: 2-3 ★)
11. Hensel's Lemma
(round: 3-4 ★, usefulness: 3 ★, difficulty: 3 ★)

12. Gauß' Lemma (irreducibility, primitivity)
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 2.5 ★)
13. Lagrange interpolation
(round: 3-4 ★, usefulness: 3.5 ★, difficulty: 3-4 ★)

2.5 other results and methods

1. telescoping sums and products
(round: 1 ★, usefulness: 3.5 ★, difficulty: 2 ★)
2. swapping sums method
(round: 2 ★, usefulness: 4 ★, difficulty: 1.5 ★)
3. Iverson Bracket, Indicator Function
(round: 2 ★, usefulness: 4 ★, difficulty: 2-3.5 ★)
4. Kronecker-Capelli's Theorem
(round: 2-3 ★, usefulness: 2 ★, difficulty: 2 ★)
5. coefficient extraction
(round: 2-3 ★, usefulness: 3 ★, difficulty: 2.5 ★)
6. Partial Fraction Decomposition
(round: 3 ★, usefulness: 2.5 ★, difficulty: 2.5 ★)
7. generating functions
(round: 3 ★, usefulness: 4 ★, difficulty: 2-4 ★)
8. Snake Oil method
(round: 3-4 ★, usefulness: 3 ★, difficulty: 2.5 ★)
9. Abel Summation
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 2.5 ★)
10. Erdos-Szekeres Theorem
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 2.5 ★)
11. integral technique
(round: 3-4 ★, usefulness: 3 ★, difficulty: 3 ★)
12. Fourier Transforms
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 3-4 ★)
13. open unit disk
(round: 3-4 ★, usefulness: ? ★, difficulty: 2.5 ★)
14. Parseval's Theorem
(round: 4 ★, usefulness: 1 ★, difficulty: 3.5 ★)
15. algebraic curves
(round: ? ★, usefulness: 1 ★, difficulty: 4+ ★)
16. Newton Polytope
(round: ? ★, usefulness: 1 ★, difficulty: 4+ ★)

17. real stable polynomials
(round: ? ★, usefulness: 1 ★, difficulty: ? ★)
18. Cesaro Mean Theorem
(round: ? ★, usefulness: 1 ★, difficulty: ? ★)

3 Number Theory

3.1 classical Number Theory

1. prime factorization
(round: 1 ★, usefulness: 3 ★, difficulty: 1 ★)
2. divisibility conditions of 2, 3, 5, 7, 11, 13; $a \mid b, b \mid a \Rightarrow a = b$
(round: 1 ★, usefulness: 3 ★, difficulty: 1-2 ★)
3. Euclidean Algorithm
(round: 2 ★, usefulness: 3 ★, difficulty: 1 ★)
4. gcd, lcm and some properties
(round: 2 ★, usefulness: 3 ★, difficulty: 1-2 ★)
5. floor and ceil function + properties
(round: 2 ★, usefulness: 4 ★, difficulty: 1-4 ★)
6. lattice point proof (identity $\sum_{k=1}^{m-1} \lfloor \frac{kn}{m} \rfloor = \frac{(m-1)(n-1) + \gcd(m,n) - 1}{2}$)
(round: 2 ★, usefulness: 3 ★, difficulty: 2 ★)
7. Four Numbers Lemma
(round: ? ★, usefulness: 1.5 ★, difficulty: 1 ★)
8. Hermite identity
(round: ? ★, usefulness: 2 ★, difficulty: 1.5 ★)
9. Bezout's Theorem
(round: 2-3 ★, usefulness: 3 ★, difficulty: 2 ★)
10. numeral systems
(round: 2-3 ★, usefulness: 2.5 ★, difficulty: 2 ★)
11. Fermat's Two Squares Theorem
(round: ? ★, usefulness: 2 ★, difficulty: 1.5 ★)
12. Lagrange's Four Squares Theorem
(round: ? ★, usefulness: 2 ★, difficulty: 2 ★)
13. Fermat numbers
(round: 2-3 ★, usefulness: 2 ★, difficulty: 2 ★)
14. Proth numbers
(round: 2-3 ★, usefulness: 1.5 ★, difficulty: 1.5 ★)
15. factorization motivated by roots of unity
(round: 2-3 ★, usefulness: 2.5 ★, difficulty: 2 ★)
16. number and sum of divisors
(round: 2-3 ★, usefulness: 3 ★, difficulty: 2 ★)
17. Vandermonde's identity
(round: 3 ★, usefulness: 3 ★, difficulty: 2 ★)
18. Faulhaber's Formula
(round: 3 ★, usefulness: 2.5 ★, difficulty: 2.5 ★)

19. Catalan's conjecture/Mihailescu's Theorem
(round: 3 ★, usefulness: 3 ★, difficulty: 2.5 ★)
20. Carmichael's Theorem
(round: ? ★, usefulness: 1.5 ★, difficulty: 3 ★)
21. Sylvester-Schur Theorem
(round: ? ★, usefulness: 1.5 ★, difficulty: 2.5 ★)
22. p -adic valuation
(round: 3-4 ★, usefulness: 4 ★, difficulty: 2.5-4 ★)
23. Legendre's formula and identity $v_p(n!) = \frac{n-s_p(n)}{p-1}$
(round: 3-4 ★, usefulness: 3.5 ★, difficulty: 2.5-4 ★)
24. LTE Lemma
(round: 3-4 ★, usefulness: 3.5 ★, difficulty: 2.5-4 ★)
25. Kummer's Theorem
(round: 3-4 ★, usefulness: 2 ★, difficulty: 3 ★)
26. Jacobi + Legendre symbol
(round: 3-4 ★, usefulness: 3.5 ★, difficulty: 2.5-4 ★)
27. identity $\sum_{n=1}^p \left(\frac{an+b}{p} \right) = 0; p > 2, \gcd(a, p) = 1$
(round: 3-4 ★, usefulness: 2 ★, difficulty: 3 ★)
28. continued fractions, diophantine approximation/Dirichlet's approximation Theorem
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 3 ★)
29. Zsigmondy's Theorem (sometimes "banned", be careful!)
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 3.5 ★)
30. Roth's Theorem
(round: 4 ★, usefulness: 1.5 ★, difficulty: 4 ★)

3.1.1 Modular Arithmetic

1. Modulo and Residue classes
(round: 1 ★, usefulness: 4 ★, difficulty: 1 ★)
2. Fermat's Little Theorem
(round: 1 ★, usefulness: 3 ★, difficulty: 1 ★)
3. Euler's Totient Function, Eulers Theorem
(round: 1 ★, usefulness: 3 ★, difficulty: 1 ★)
4. Wilson's Theorem
(round: 2 ★, usefulness: 2 ★, difficulty: 1 ★)
5. multiplicative inverse
(round: 2 ★, usefulness: 3 ★, difficulty: 1.5 ★)
6. identity $\left[\frac{x}{p} \right] = \frac{x-r}{p}, x \equiv r \pmod{p}$
(round: 2 ★, usefulness: 2.5 ★, difficulty: 2 ★)

7. Wolstenholme's Theorem
(round: ? ★, usefulness: 1.5 ★, difficulty: 2.5 ★)
8. Freshman's dream ($(x + y)^p \equiv x^p + y^p \pmod{p}$)
(round: 2-3 ★, usefulness: 2 ★, difficulty: 2 ★)
9. orders
(round: 2-3 ★, usefulness: 2.5 ★, difficulty: 2 ★)
10. primitive root
(round: 2-3 ★, usefulness: 2.5 ★, difficulty: 2 ★)
11. Dirichlet's Theorem
(round: 3 ★, usefulness: 3.5 ★, difficulty: 2 ★)
12. Lucas' Theorem
(round: 3 ★, usefulness: 2 ★, difficulty: 2.5 ★)
13. quadratic residues. Quadratic reciprocity Theorem, Eulers criterion
(round: 3 ★, usefulness: 3.5 ★, difficulty: 2.5-4 ★)
14. number of solutions to $x^2 \equiv a \pmod{p}$ and $x^2 + y^2 \equiv a \pmod{p}$
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 2.5 ★)
15. Frobenius coin problem/Chicken McNugget Theorem
(round: 3-4 ★, usefulness: 2 ★, difficulty: 2.5 ★)
16. Chinese Remainder Theorem
(round: 3-4 ★, usefulness: 3 ★, difficulty: 2.5 ★)
17. Thue's Lemma
(round: 3-4 ★, usefulness: 2 ★, difficulty: 2.5 ★)

3.1.2 Diophantine equations

1. bounding using inequalities
(round: 1 ★, usefulness: 4 ★, difficulty: 1-4 ★)
2. parity argument
(round: 1 ★, usefulness: 2.5 ★, difficulty: 1 ★)
3. considering modulo
(round: 1 ★, usefulness: 4 ★, difficulty: 1-2.5 ★)
4. factorization
(round: 1 ★, usefulness: 2.5 ★, difficulty: 1.5 ★)
5. substitution
(round: 1-2 ★, usefulness: 3 ★, difficulty: 2 ★)
6. Pythagorean triples
(round: 2 ★, usefulness: 2 ★, difficulty: 2 ★)
7. Pell's equation
(round: 2 ★, usefulness: 3 ★, difficulty: 2 ★)

8. parametric method
(round: 2 ★, usefulness: 2 ★, difficulty: 2.5-3.5 ★)
9. Infinite Descent
(round: 3 ★, usefulness: 3 ★, difficulty: 3 ★)
10. considering p-adic valuation
(round: 3 ★, usefulness: 4 ★, difficulty: 2.5-4 ★)
11. Vieta Jumping
(round: 3-4 ★, usefulness: 3 ★, difficulty: 3-4 ★)
12. Fermat's Last Theorem (sometimes "banned", be careful!)
(round: 3-4 ★, usefulness: 2 ★, difficulty: 3 ★)
13. methods from abstract algebra (Gaussian integers ...)
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 3-4 ★)

3.2 Analytic Number Theory

1. Big O Notation
(round: 3 ★, usefulness: 3 ★, difficulty: 3-4 ★)
2. Kuzmin-Landau Theorem
(round: 3-4 ★, usefulness: 1.5 ★, difficulty: 3.5 ★)
3. Rosser's Theorem
(round: 3-4 ★, usefulness: 1.5 ★, difficulty: 3 ★)
4. natural density
(round: 3-4 ★, usefulness: 2 ★, difficulty: 3.5 ★)
5. Merten's Theorems
(round: 3-4 ★, usefulness: 2 ★, difficulty: 3 ★)
6. Prime Number Theorem
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 3 ★)
7. (weak) Stirling approximation
(round: 3-4 ★, usefulness: 2 ★, difficulty: 3 ★)
8. $\sum_{p \in P} \frac{1}{p^2} < \frac{1}{2}$
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 3 ★)
9. Mangoldt function
(round: 4 ★, usefulness: 1.5 ★, difficulty: 4 ★)
10. Gallagher sieve
(round: 4 ★, usefulness: 1.5 ★, difficulty: 4 ★)
11. Dirichlet Density Theorem
(round: 4 ★, usefulness: 1 ★, difficulty: 3-4 ★)
12. Chebotarev Density Theorem
(round: 4 ★, usefulness: 1 ★, difficulty: 3-4 ★)

13. Kronecker's Approximation Theorem
(round: 4 ★, usefulness: 2 ★, difficulty: 3-4 ★)
14. equidistribution, Weyl's Equidistribution Theorem
(round: 4 ★, usefulness: 2.5 ★, difficulty: 4 ★)
15. Hardy-Ramanujan
(round: ? ★, usefulness: 1 ★, difficulty: 4 ★)
16. Polya-Vinogradov Theorem
(round: ? ★, usefulness: 1 ★, difficulty: 4 ★)

3.3 Algebraic Number Theory

1. fundamental theorem of symmetric polynomials
(round: 3 ★, usefulness: 2 ★, difficulty: 3 ★)
2. cyclotomic polynomials
(round: 3 ★, usefulness: 2.5 ★, difficulty: 3 ★)
3. algebraic numbers and integers
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 3-4 ★)
4. groups, fields, rings, (integral) domain, monoid
(round: 3-4 ★, usefulness: 2 ★, difficulty: 3-4 ★)
5. morphism, isomorphism, homomorphism
(round: 3-4 ★, usefulness: 1.5 ★, difficulty: 3-4 ★)
6. absolute norm, units, prime elements ...
(round: 3-4 ★, usefulness: 2 ★, difficulty: 3 ★)
7. unique factorization domain
(round: 3-4 ★, usefulness: 3 ★, difficulty: 3-4 ★)
8. Gaußian integers (+ Eisenstein integers, Hurwitz integers)
(round: 3-4 ★, usefulness: 3 ★, difficulty: 3-4 ★)
9. ideals
(round: ? ★, usefulness: 1 ★, difficulty: 3-4 ★)

3.4 other results and methods

1. Fibonacci numbers, Catalan numbers + Dyck path
(round: 1-2 ★, usefulness: 3 ★, difficulty: 2 ★)
2. Schinzel's Theorem
(round: 2-3 ★, usefulness: 2 ★, difficulty: 2 ★)
3. Pick's Theorem
(round: 2-3 ★, usefulness: 2 ★, difficulty: 2 ★)
4. Minkowski's Theorem (lattice)
(round: ? ★, usefulness: 1.5 ★, difficulty: 3 ★)
5. Farey Sequence
(round: ? ★, usefulness: 1.5 ★, difficulty: 3 ★)

6. Kobayashi's Theorem
(round: 3-4 ★, usefulness: 3 ★, difficulty: 3 ★)
7. Bertrand's postulate
(round: 3-4 ★, usefulness: 3 ★, difficulty: 3 ★)
8. Erdos-Ginsburg-Ziv Theorem
(round: 3-4 ★, usefulness: 3 ★, difficulty: 3 ★)
9. Möbius function, Möbius inversion, Dirichlet convolution
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 3-4 ★)

4 Geometry

4.1 Plane Geometry

1. area formulas $\triangle$ (Heron, $\frac{abc}{4R}$, $\frac{1}{2} \cdot ab \sin \gamma$, sr , $\frac{1}{2}a \cdot h_a$, $r_{A\text{-exradius}}(s - a)$)
(round: 1 ★, usefulness: 4 ★, difficulty: 1 ★)
2. Pythagoras' Theorem + related
(round: 1 ★, usefulness: 2.5 ★, difficulty: 1 ★)
3. similarity, congruency
(round: 1 ★, usefulness: 4 ★, difficulty: 1-4 ★)
4. incircle, circumcircle, excircle + lemmas
(round: 1 ★, usefulness: 4 ★, difficulty: 2 ★)
5. angles when dealing with circles
(round: 1 ★, usefulness: 4 ★, difficulty: 1-3 ★)
6. angle chasing + directed angles
(round: 1 ★, usefulness: 4 ★, difficulty: 1-4 ★)
7. incenter, circumcenter, excenter, centroid, orthocentre + lemmas
(round: 1 ★, usefulness: 4 ★, difficulty: 2 ★)
8. trigonometry (extended law of sines, ...)
(round: 1 ★, usefulness: 3.5 ★, difficulty: 2 ★)
9. extend lines, draw line between given points, parallelogram trick
(round: 1 ★, usefulness: 4 ★, difficulty: 1-3 ★)
10. Finsler-Hadwiger Theorem
(round: ? ★, usefulness: 1.5 ★, difficulty: 1 ★)
11. Varignon's Theorem
(round: 1-2 ★, usefulness: 2.5 ★, difficulty: 1 ★)
12. Phantom points (define X' , show $X' \equiv X$)
(round: 2 ★, usefulness: 4 ★, difficulty: 2-4 ★)
13. Ceva's and Menelaus' Theorem
(round: 2 ★, usefulness: 4 ★, difficulty: 2-3 ★)
14. Stewart's Theorem
(round: 2 ★, usefulness: 2.5 ★, difficulty: 2 ★)
15. van Aubel's Theorem + Petr-Douglas-Neumann Theorem (+ Napoleon's Theorem)
(round: 2 ★, usefulness: 3 ★, difficulty: 2 ★)
16. Ptolemy's Theorem
(round: 2 ★, usefulness: 2.5 ★, difficulty: 2 ★)
17. Angle Bisector Theorem + Ratio Lemma
(round: 2 ★, usefulness: 3.5 ★, difficulty: 2 ★)
18. Viviani's Theorem
(round: 2 ★, usefulness: 2 ★, difficulty: 2 ★)

19. British Flag Theorem
(round: ? ★, usefulness: 2 ★, difficulty: 1.5 ★)
20. Steiner-Lehmus Theorem
(round: ? ★, usefulness: 1.5 ★, difficulty: 2 ★)
21. van Schooten's Theorem
(round: ? ★, usefulness: 2 ★, difficulty: 2 ★)
22. symmedian + lemmas
(round: 2-3 ★, usefulness: 3 ★, difficulty: 2.5 ★)
23. Simson line Theorem
(round: 2-3 ★, usefulness: 2 ★, difficulty: 2.5 ★)
24. Three Tangents Lemma
(round: 2-3 ★, usefulness: 2 ★, difficulty: 2 ★)
25. Bretschneider's Formula
(round: 2-3 ★, usefulness: 2 ★, difficulty: 2 ★)
26. Kosnita's Theorem
(round: ? ★, usefulness: 2 ★, difficulty: 2.5 ★)
27. Carnot's Theorem (inradius, circumradius), (conics), (perpendicularity)
(round: ? ★, usefulness: 2.5 ★, difficulty: 2.5 ★)
28. Japanese Theorem for cyclic polygons
(round: ? ★, usefulness: 2.5 ★, difficulty: 2.5 ★)
29. Descartes' Theorem
(round: ? ★, usefulness: 1 ★, difficulty: 2.5 ★)
30. Euler's Theorem circumradius-inradius
(round: ? ★, usefulness: 2 ★, difficulty: 2.5 ★)
31. Desargue's Theorem
(round: 3 ★, usefulness: 3 ★, difficulty: 3 ★)
32. Euler line
(round: 3 ★, usefulness: 3.5 ★, difficulty: 3 ★)
33. de Longchamps point, Exeter point, Fermat point, Gergonne point, Lemoine point
(round: 3 ★, usefulness: 2 ★, difficulty: 2.5 ★)
34. Bevan point, Nagel point, Miquel point + Theorem
(round: 3 ★, usefulness: 3 ★, difficulty: 3 ★)
35. nine point circle, eight point circle
(round: 3 ★, usefulness: 2.5 ★, difficulty: 3 ★)
36. isotomic + isogonal conjugate
(round: 3 ★, usefulness: 2.5 ★, difficulty: 3 ★)
37. power of a point
(round: 3 ★, usefulness: 3.5 ★, difficulty: 3-4 ★)

- 38. homothety
(round: 3 ★, usefulness: 3.5 ★, difficulty: 2-4 ★)
- 39. spiral similarity
(round: 3 ★, usefulness: 3 ★, difficulty: 3-4 ★)
- 40. projective geometry (cross-ratio, La Hire's Theorem, Brokard's Theorem) + lemmas
(round: 3 ★, usefulness: 3.5 ★, difficulty: 2.5-4 ★)
- 41. inversion, $\sqrt{bc}$ -inversion, $\sqrt{\frac{bc}{2}}$ -inversion
(round: 3 ★, usefulness: 3 ★, difficulty: 3-4 ★)
- 42. Apollonius' circle
(round: ? ★, usefulness: 2 ★, difficulty: 2.5 ★)
- 43. Casey's Theorem
(round: ? ★, usefulness: 1.5 ★, difficulty: 3 ★)
- 44. complete quadrilateral
(round: 3-4 ★, usefulness: 3 ★, difficulty: 3 ★)
- 45. Newton-Gauß line, ((Anti-)Steiner line, Collings Clime, Shatunov-Tokarev line, ...)
(round: 3-4 ★, usefulness: 2 ★, difficulty: 3-4 ★)
- 46. Humpty point, Dumpty point, Sharkydevil point, Queue point
(round: 3-4 ★, usefulness: 3 ★, difficulty: 3-4 ★)
- 47. Death Star Lemma
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 3-4 ★)
- 48. Salmon Lemma
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 3-4 ★)
- 49. Brianchon's Theorem and Pascal's Theorem
(round: 3-4 ★, usefulness: 3 ★, difficulty: 3-4 ★)
- 50. Pappus' Theorem
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 3-4 ★)
- 51. Butterfly Theorem
(round: 3-4 ★, usefulness: 2 ★, difficulty: 3-4 ★)
- 52. curvilinear incircle
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 3-4 ★)
- 53. mixtilinear incircle
(round: 3-4 ★, usefulness: 3.5 ★, difficulty: 3-4 ★)
- 54. insimilicenter (X_{55}), exsimilicenter (X_{56})
(round: 4 ★, usefulness: 2.5 ★, difficulty: 3-4 ★)
- 55. Iran Lemma
(round: 4 ★, usefulness: 2 ★, difficulty: 3-4 ★)
- 56. Sondat's Theorem
(round: 4 ★, usefulness: 1.5 ★, difficulty: 3-4 ★)

- 57. minimal model program, (un-)tethered moving points, projective transformations, degree counting
(round: 4 ★, usefulness: 3 ★, difficulty: 4 ★)
- 58. involution, Desargue's involution Theorem, Dual of Desargue's involution Theorem
(round: 4 ★, usefulness: 2.5 ★, difficulty: 4 ★)
- 59. Monge-D'Alembert's Theorem
(round: 4 ★, usefulness: 2 ★, difficulty: 3-4 ★)
- 60. cubics, (rectangular) hyperbolas
(round: 4 ★, usefulness: 2 ★, difficulty: 4 ★)
- 61. permutation lines
(round: 4 ★, usefulness: 2 ★, difficulty: 4 ★)
- 62. Bezout's Theorem (algebraic geometry version)
(round: 4 ★, usefulness: 2 ★, difficulty: 3.5 ★)
- 63. Cayley Bacharach
(round: 4 ★, usefulness: 2 ★, difficulty: 4 ★)
- 64. Kiepert conics, Jerabek hyperbola, Artzt parabolas
(round: 4 ★, usefulness: 2 ★, difficulty: 4 ★)
- 65. Dehn-Invariant
(round: 4 ★, usefulness: 1 ★, difficulty: 4 ★)

4.2 Solid Geometry

- 1. simplification to two dimensions
(round: 1 ★, usefulness: 4 ★, difficulty: 1-3 ★)
- 2. area, volume and length formulas
(round: 1 ★, usefulness: 4 ★, difficulty: 1 ★)
- 3. Euler's characteristic
(round: ? ★, usefulness: 1.5 ★, difficulty: 1 ★)
- 4. De Gua's Theorem
(round: 2 ★, usefulness: 1.5 ★, difficulty: 1.5 ★)
- 5. Commandino's Theorem
(round: ? ★, usefulness: 2 ★, difficulty: 2 ★)
- 6. stereographic projection
(round: 3 ★, usefulness: 3 ★, difficulty: 3 ★)
- 7. Dandelin sphere
(round: 3 ★, usefulness: 2.5 ★, difficulty: 3 ★)

4.3 Analytic Geometry

(formulas for $\perp, \parallel, \circ, \cap, \angle \dots$)

1. trigonometry bash + formulas
(round: 1 ★, usefulness: 3 ★, difficulty: 1-4 ★)
2. cartesian coordinates and vector bash + formulas
(round: 2 ★, usefulness: 2.5 ★, difficulty: 2-3 ★)
3. complex coordinates + formulas
(round: 2-3 ★, usefulness: 4 ★, difficulty: 2-4 ★)
4. barycentric coordinates + formulas
(round: 3 ★, usefulness: 3 ★, difficulty: 3-4 ★)
5. trilinear coordinates + formulas
(round: 3-4 ★, usefulness: 2 ★, difficulty: 3-4 ★)

5 Combinatorics

5.1 basic methods

1. basic formulas (binomial coefficients + identities, factorial, ...)
(round: 1 ★, usefulness: 4 ★, difficulty: 1 ★)
2. pigeonhole principle
(round: 1 ★, usefulness: 3 ★, difficulty: 1.5 ★)
3. principle of inclusion and exclusion
(round: 1-2 ★, usefulness: 3 ★, difficulty: 2 ★)
4. recursion (+ linear recursion formula)
(round: 2 ★, usefulness: 3 ★, difficulty: 2-3 ★)
5. double counting
(round: 2 ★, usefulness: 4 ★, difficulty: 2-4 ★)
6. injective, surjective, bijective mapping
(round: 2-3 ★, usefulness: 3.5 ★, difficulty: 2-4 ★)
7. local adjustment method
(round: 2-3 ★, usefulness: 3 ★, difficulty: 3 ★)
8. generating functions, exponential generating functions
(round: 3 ★, usefulness: 4 ★, difficulty: 3-4 ★)

5.2 Graph Theory

1. maximum length of path
(round: 1-2 ★, usefulness: 2.5 ★, difficulty: 1.5 ★)
2. graph coloring
(round: 2 ★, usefulness: 4 ★, difficulty: 2-3.5 ★)
3. induced subgraph
(round: 2-3 ★, usefulness: 2 ★, difficulty: 2 ★)
4. complement graph
(round: 2-3 ★, usefulness: 3.5 ★, difficulty: 3 ★)
5. partition graph, bipartite graph
(round: 2-3 ★, usefulness: 3 ★, difficulty: 3-4 ★)
6. cycles, Hamiltonian cycle, Eulerian cycle (+ Hierholzer's Theorem)
(round: 2-3 ★, usefulness: 3 ★, difficulty: 3 ★)
7. clique
(round: 2-3 ★, usefulness: 3.5 ★, difficulty: 3 ★)
8. dominating set
(round: 3 ★, usefulness: 2 ★, difficulty: 2.5 ★)
9. tree, spanning tree, isomorphic tree
(round: 3 ★, usefulness: 3.5 ★, difficulty: 2-3 ★)

10. tournaments
(round: 3 ★, usefulness: 3 ★, difficulty: 3 ★)
11. radius and diameter of a graph
(round: 3 ★, usefulness: 2 ★, difficulty: 3 ★)
12. Ramsey's Theorem, Hypergraph Ramsey
(round: 3 ★, usefulness: 4 ★, difficulty: 2-4 ★)
13. Hall's Marriage Theorem
(round: 3 ★, usefulness: 3.5 ★, difficulty: 3 ★)
14. Turan's Theorem
(round: 3 ★, usefulness: 4 ★, difficulty: 3 ★)
15. BFS and DFS algorithm
(round: 3 ★, usefulness: 3 ★, difficulty: 2.5 ★)
16. k-connected graph, block graph
(round: 3-4 ★, usefulness: 3 ★, difficulty: 3 ★)
17. bridges
(round: 3-4 ★, usefulness: 2 ★, difficulty: 3.5 ★)
18. centroid of a tree
(round: 3-4 ★, usefulness: 2 ★, difficulty: 3.5 ★)
19. rim (chordless cycle)
(round: 3-4 ★, usefulness: 2 ★, difficulty: 3.5 ★)
20. directed acyclic graph, topological ordering
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 3-4 ★)
21. König's Theorem
(round: 3-4 ★, usefulness: 3 ★, difficulty: 3.5 ★)
22. planar graphs, Kuratowski's Theorem
(round: ? ★, usefulness: 2 ★, difficulty: 3-4 ★)
23. degree sequence
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 3-4 ★)
24. Kundu's k-factor Theorem
(round: 3-4 ★, usefulness: 3 ★, difficulty: 3.5 ★)
25. Menger's Theorem
(round: 3-4 ★, usefulness: 2 ★, difficulty: 3.5 ★)
26. Bondy-Chvatal (+ Dirac, Ore, Ghouila-Houriri, Meyniel, Rahman-Kaykobad's Theorems)
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 3-4 ★)
27. Dilworth's Theorem
(round: 3-4 ★, usefulness: 2 ★, difficulty: 3-4 ★)
28. Szekeres-Wilf Theorem
(round: ? ★, usefulness: 1 ★, difficulty: ? ★)

- 29. Moon-Moser's Theorem
(round: ? ★, usefulness: 1 ★, difficulty: ? ★)
- 30. Graham-Pollack's Theorem
(round: ? ★, usefulness: 2.5 ★, difficulty: 3-4 ★)
- 31. Bregman-Minc's inequality, Szele's Theorem, Caro-Wei Theorem, Erdos-Ko-Rado Theorem
(round: ? ★, usefulness: 1 ★, difficulty: 4 ★)
- 32. score sequence, Landau's Theorem
(round: ? ★, usefulness: 2.5 ★, difficulty: 3-4 ★)
- 33. Gallai-Hasse-Roy-Vitaver Theorem
(round: 4 ★, usefulness: 2 ★, difficulty: 3-4 ★)
- 34. Kendall tau rank distance / bubble sort distance
(round: 4 ★, usefulness: 2 ★, difficulty: 3-4 ★)
- 35. median order of tournament
(round: ? ★, usefulness: 2.5 ★, difficulty: 3-4 ★)
- 36. matroid
(round: ? ★, usefulness: 1 ★, difficulty: 4+ ★)

5.3 Sets

- 1. double counting
(round: 2 ★, usefulness: 4 ★, difficulty: 2-4 ★)
- 2. roots of unity filter
(round: 3 ★, usefulness: 3.5 ★, difficulty: 3-4 ★)
- 3. Hall's Marriage Theorem
(round: 3-4 ★, usefulness: 3 ★, difficulty: 3-4 ★)
- 4. Radon's Theorem
(round: ? ★, usefulness: 1 ★, difficulty: 3-4 ★)
- 5. Helly's Theorem
(round: 3-4 ★, usefulness: 2 ★, difficulty: 3-4 ★)
- 6. Lubell-Yamamoto-Meshalkin inequality
(round: 3-4 ★, usefulness: 1.5 ★, difficulty: 3-4 ★)
- 7. Minkowski sum
(round: ? ★, usefulness: 1.5 ★, difficulty: ? ★)
- 8. Cauchy-Davenport Theorem
(round: 4 ★, usefulness: 2.5 ★, difficulty: 3.5 ★)
- 9. Kneser's Theorem
(round: 4 ★, usefulness: 2 ★, difficulty: 3.5 ★)
- 10. Sperner's Theorem
(round: 4 ★, usefulness: 2.5 ★, difficulty: 4 ★)

11. Mann's Theorem
(round: ? ★, usefulness: 1.5 ★, difficulty: 3.5 ★)
12. Kleitmann's Lemma
(round: 4 ★, usefulness: 2 ★, difficulty: 4 ★)
13. Frankl's conjecture (bound $\frac{n-1}{\log_2 n}$)
(round: 4 ★, usefulness: 1 ★, difficulty: 3.5 ★)

5.4 Games, Tiling, Algorithms

1. Zermelo's Theorem
(round: 1 ★, usefulness: 2 ★, difficulty: 1 ★)
2. block walking
(round: 1 ★, usefulness: 1.5 ★, difficulty: 1 ★)
3. board coloring to reach contradiction
(round: 1 ★, usefulness: 3 ★, difficulty: 2 ★)
4. copy opponent's move, adapt to opponent's move
(round: 1 ★, usefulness: 4 ★, difficulty: 1-4 ★)
5. construct algorithm
(round: 1 ★, usefulness: 4 ★, difficulty: 1-4 ★)
6. greedy algorithm
(round: 1-2 ★, usefulness: 2.5 ★, difficulty: 1-3 ★)
7. gas station problem, Raney's Theorem
(round: 2 ★, usefulness: 3 ★, difficulty: 1.5-4 ★)
8. numbering tiles (e.g. ± 1)
(round: 2 ★, usefulness: 3 ★, difficulty: 2.5 ★)
9. invariants, monovariants
(round: 2 ★, usefulness: 4 ★, difficulty: 1-3 ★)
10. polynomials method + roots of unity for tiling
(round: 3 ★, usefulness: 3 ★, difficulty: 3 ★)

5.5 other results and methods

1. Sylvester-Gallai Theorem, De Bruijn-Erdos Theorem
(round: 2 ★, usefulness: 2 ★, difficulty: 2 ★)
2. Combinatorial Nullstellensatz
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 3-4 ★)
3. Hook Length Formula and Young Tableaux
(round: 4 ★, usefulness: 2 ★, difficulty: 3.5 ★)
4. Allowable Sequences
(round: ? ★, usefulness: 1 ★, difficulty: ? ★)

6 Probability

1. expected value, linearity of expectation
(round: 2 ★, usefulness: 3.5 ★, difficulty: 2-4 ★)
2. variance, covariance, standard deviation
(round: ? ★, usefulness: 2 ★, difficulty: 3 ★)
3. Probabilistic Method, moment method
(round: 3-4 ★, usefulness: 4 ★, difficulty: 4 ★)
4. Markov's inequality
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 3-4 ★)
5. Chebyshev's inequality
(round: 3-4 ★, usefulness: 2 ★, difficulty: 3-4 ★)
6. Paley-Zygmund inequality
(round: 3-4 ★, usefulness: 2 ★, difficulty: 3-4 ★)
7. moment generating function
(round: ? ★, usefulness: 1.5 ★, difficulty: 4 ★)
8. Chernoff bound
(round: 4 ★, usefulness: 2 ★, difficulty: 4 ★)
9. Lovasz Local Lemma
(round: 4 ★, usefulness: 3 ★, difficulty: 4 ★)

7 Linear Algebra

1. Kronecker-Capelli's Theorem
(round: 3 ★, usefulness: 2.5 ★, difficulty: 2.5 ★)
2. incidence matrix
(round: 3 ★, usefulness: 2.5 ★, difficulty: 2.5 ★)
3. Fischer's inequality
(round: ? ★, usefulness: 1 ★, difficulty: 3-4 ★)
4. Vandermonde matrix
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 3-4 ★)
5. Hadamard inequality
(round: 3-4 ★, usefulness: 2.5 ★, difficulty: 3-4 ★)
6. Hamel basis
(round: 4 ★, usefulness: 2 ★, difficulty: 4 ★)

8 resources for self-study

8.1 books

- Modern Olympiad Number Theory - Aditya Khurmi
- 104 Number Theory Problems - Titu Andreescu, Dorin Andrica, Zuming Feng
- An Introduction to Diophantine Equations - T. Andreescu, D. Andrica, I. Cucurezeanu
- Analytic Number Theorey - Donald J. Newman
- An Introduction to Algebraic Number Theory through Olympiad Problems - Elias Caeiro (download on AOPS)

- Combinatorial Problems in Mathematical Competitions - Yao Zhang
- Graph Theory - Liu Ruifang, Zhai Mingqing, Lin Yuanqing
- Olympiad Combinatorics - Pranav A. Sriram
- Graph Theory- Reinhard Diestel
- The Probabilistic Method - Noga Alon, Joel H. Spencer

- Euclidean Geometry in Mathematical Olympiads - Evan Chen
- Solving Problems in Geometry, Insights and Strategies for Mathematical Olympiad and Competitions - Kim Hoo Hang, Haibin Wang
- Modern Pure Solid Geometry - Nathan Altshiller-Court

- Functional Equation - Pang-Cheng Wu
- Algebraic Inequalities - Vasile Cirtoaje
- Inequalities, A Mathematical Olympiad Approach - R.B. Manfrino, J.A.G. Ortega, R.V. Delgado

- Problems From The Book - Titu Andreescu, Gabriel Dospinescu
- Straight From The Book - Titu Andreescu, Gabriel Dospinescu
- Problem Solving Strategies - Arthur Engel

8.2 internet and blogs

- <https://artofproblemsolving.com/>
(AOPS)
- <https://www.cut-the-knot.org/>
(cut the knot)
- <https://web.evanchen.cc/> and <https://blog.evanchen.cc/all-posts/>
(Evan Chen's blogs)
- <https://yufeizhao.com/olympiad/>
(Yufei Zhao's handouts)
- <https://artofproblemsolving.com/community/c3735646>
(EthanWYX' blog)
- <https://artofproblemsolving.com/community/c1824893>
(MathsJames' blog)
- <https://artofproblemsolving.com/community/c284651>
(Telv Cohl's blog)
- <https://artofproblemsolving.com/community/c3739355h3359335>
(kingu's blog)
- <https://artofproblemsolving.com/community/c1941956>
(CANBANKAN's blog)
- <https://artofproblemsolving.com/community/c1140984>
(YaoAOPS' blog)
- <https://dgrozev.wordpress.com/>
(Dragomir Grozev's blog)

8.3 handouts and articles

- Probabilistic Method in Combinatorics - Po-Shen Lo
- Sledgehammers in Number Theory - Carl Joshua Quines
- A Taste of Analytic Number Theory - Ayan Nath
- Density and Complexity - Eric Shen
- Nuclear Geometry - Eric Shen
- Obscure Geometry Theorem - Carl Joshua Quines
- Mixtilinear circle in olympiad geometry - Dogu, Paramizo Dicrominique, Dsoong
- Inequalities Cheatsheet - Laszlo Kozma
- Evan Chen's handouts (see above)
- Yufei Zhao's handouts (see above)